

FIFA World Cup Analysis

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GitHub Repository:

<https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis>

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Question 1: Data Inspection, Cleaning and Home Penalty Distribution

Rationale

The four datasets were joined so that match, team, stadium and tournament information could be analysed together. Missing values "?" were changed to NA. The variables **Result**, **Stage**, **Country** and **ExtraTime** were changed to factors because they contain categories. **HomePenalty** and **AwayPenalty** were changed to numeric variables because they contain numbers of penalty goals.

A bar chart was chosen because **HomePenalty** contains distinct integer values. The height of each bar shows how many penalty shootouts had penalty scores for each home team.

Load and Join the Data

```
# Read the data CSV files.
matches <- read_csv("../data/Matches.csv")
stadiums <- read_csv("../data/Stadiums.csv")
teams <- read_csv("../data/Teams.csv")
tournaments <- read_csv("../data/Tournaments.csv")

# Change every "?" value to NA.
matches[matches == "?"] <- NA
stadiums[stadiums == "?"] <- NA
teams[teams == "?"] <- NA
tournaments[tournaments == "?"] <- NA

# Join team information for the home team and away team.
df <- matches %>%
  left_join(
    teams %>%
      rename(
        HomeTeamID = TeamID,
        HomeTeamName = TeamName,
        HomeTeamCode = TeamCode
      ),
    by = "HomeTeamID"
  ) %>%
  left_join(
    teams %>%
      rename(
        AwayTeamID = TeamID,
        AwayTeamName = TeamName,
        AwayTeamCode = TeamCode
      ),
    by = "AwayTeamID"
  ) %>%
  left_join(stadiums, by = "StadiumID") %>%
  left_join(tournaments, by = "TournamentID")
```

Inspect and Clean the Data

```
# Display the first six rows.
head(df)
```

```
## # A tibble: 6 x 24
##   MatchID TournamentID MatchName Stage MatchDate MatchTime StadiumID HomeTeamID
##   <chr>      <chr>      <chr>    <chr> <chr>      <time>      <chr>      <chr>
## 1 M-1930-- WC-1930      France v~ grou~ 7/13/1930 15:00      S-193      T-28
## 2 M-1930-- WC-1930      United S~ grou~ 7/13/1930 15:00      S-192      T-80
## 3 M-1930-- WC-1930      Yugoslav~ grou~ 7/14/1930 12:45      S-192      T-84
## 4 M-1930-- WC-1930      Romania ~ grou~ 7/14/1930 14:50      S-193      T-59
## 5 M-1930-- WC-1930      Argentin~ grou~ 7/15/1930 16:00      S-192      T-03
## 6 M-1930-- WC-1930      Chile v ~ grou~ 7/16/1930 14:45      S-192      T-13
## # i 16 more variables: AwayTeamID <chr>, HomeTeamScore <dbl>,
## #   AwayTeamScore <dbl>, ExtraTime <dbl>, PenaltyShootout <dbl>,
## #   HomePenalty <chr>, AwayPenalty <chr>, Result <chr>, HomeTeamName <chr>,
## #   HomeTeamCode <chr>, AwayTeamName <chr>, AwayTeamCode <chr>,
## #   StadiumName <chr>, City <chr>, Country <chr>, TournamentName <chr>
```

```
# Check the number of rows and columns.
dim(df)
```

```
## [1] 964 24
```

```
# Check the variable types.
str(df)
```

```
## spc_tbl_ [964 x 24] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ MatchID      : chr [1:964] "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr [1:964] "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName     : chr [1:964] "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "R
## $ Stage         : chr [1:964] "group stage" "group stage" "group stage" "group stage" ...
## $ MatchDate     : chr [1:964] "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime     : 'hms' num [1:964] 15:00:00 15:00:00 12:45:00 14:50:00 ...
##   .. attr(*, "units")= chr "secs"
## $ StadiumID     : chr [1:964] "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID    : chr [1:964] "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID    : chr [1:964] "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : num [1:964] 4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : num [1:964] 1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime     : num [1:964] 0 0 0 0 0 0 0 0 0 0 ...
## $ PenaltyShootout: num [1:964] 0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty   : chr [1:964] NA "0" "0" "0" ...
## $ AwayPenalty   : chr [1:964] "0" "0" "0" NA ...
## $ Result        : chr [1:964] "home team win" "home team win" "home team win" "home team win" ...
## $ HomeTeamName  : chr [1:964] "France" "United States" "Yugoslavia" "Romania" ...
## $ HomeTeamCode  : chr [1:964] "FRA" "USA" "YUG" "ROU" ...
## $ AwayTeamName  : chr [1:964] "Mexico" "Belgium" "Brazil" "Peru" ...
## $ AwayTeamCode  : chr [1:964] "MEX" "BEL" "BRA" "PER" ...
## $ StadiumName   : chr [1:964] "Estadio Pocitos" "Estadio Gran Parque Central" "Estadio Gran Parque
```

```
## $ City          : chr [1:964] "Montevideo" "Montevideo" "Montevideo" "Montevideo" ...
## $ Country       : chr [1:964] "Uruguay" "Uruguay" "Uruguay" "Uruguay" ...
## $ TournamentName : chr [1:964] "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIFA World Cup" ...
## - attr(*, "spec")=
## .. cols(
## ..   MatchID = col_character(),
## ..   TournamentID = col_character(),
## ..   MatchName = col_character(),
## ..   Stage = col_character(),
## ..   MatchDate = col_character(),
## ..   MatchTime = col_time(format = ""),
## ..   StadiumID = col_character(),
## ..   HomeTeamID = col_character(),
## ..   AwayTeamID = col_character(),
## ..   HomeTeamScore = col_double(),
## ..   AwayTeamScore = col_double(),
## ..   ExtraTime = col_double(),
## ..   PenaltyShootout = col_double(),
## ..   HomePenalty = col_character(),
## ..   AwayPenalty = col_character(),
## ..   Result = col_character()
## .. )
## - attr(*, "problems")=<externalptr>
```

Display a summary of the main variables.

```
summary(
  df[c(
    "Result",
    "Stage",
    "Country",
    "ExtraTime",
    "PenaltyShootout",
    "HomePenalty",
    "AwayPenalty"
  )]
)
```

```
##      Result          Stage          Country          ExtraTime
## Length:964      Length:964      Length:964      Min.   :0.00000
## Class :character Class :character Class :character 1st Qu.:0.00000
## Mode  :character Mode  :character Mode  :character Median :0.00000
##                                         Mean  :0.07573
##                                         3rd Qu.:0.00000
##                                         Max.   :1.00000
## PenaltyShootout HomePenalty      AwayPenalty
## Min.   :0.00000      Length:964      Length:964
## 1st Qu.:0.00000      Class :character Class :character
## Median :0.00000      Mode  :character Mode  :character
## Mean   :0.03631
## 3rd Qu.:0.00000
## Max.   :1.00000
```

```

# Convert categorical variables to factors.
df$Result <- as.factor(df$Result)
df$Stage <- as.factor(df$Stage)
df$Country <- as.factor(df$Country)
df$ExtraTime <- as.factor(df$ExtraTime)

# Convert penalty variables to numeric.
df$HomePenalty <- as.numeric(df$HomePenalty)
df$AwayPenalty <- as.numeric(df$AwayPenalty)

# Replace missing penalty values with zero.
df$HomePenalty[is.na(df$HomePenalty)] <- 0
df$AwayPenalty[is.na(df$AwayPenalty)] <- 0

```

Code Testing

```

# The number of rows should be unchanged after joining.
nrow(matches)

```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```

# Check the classes after conversion.
class(df$Result)

```

```
## [1] "factor"
```

```
class(df$Stage)
```

```
## [1] "factor"
```

```
class(df$Country)
```

```
## [1] "factor"
```

```
class(df$ExtraTime)
```

```
## [1] "factor"
```

```
class(df$HomePenalty)
```

```
## [1] "numeric"
```

```
class(df$AwayPenalty)
```

```
## [1] "numeric"
```

```
# Check that no missing penalty values remain.  
sum(is.na(df$HomePenalty))
```

```
## [1] 0
```

```
sum(is.na(df$AwayPenalty))
```

```
## [1] 0
```

The two row counts should be equal. The four categorical variables should return "factor", the two penalty variables should return "numeric", and both missing-value checks should return zero.

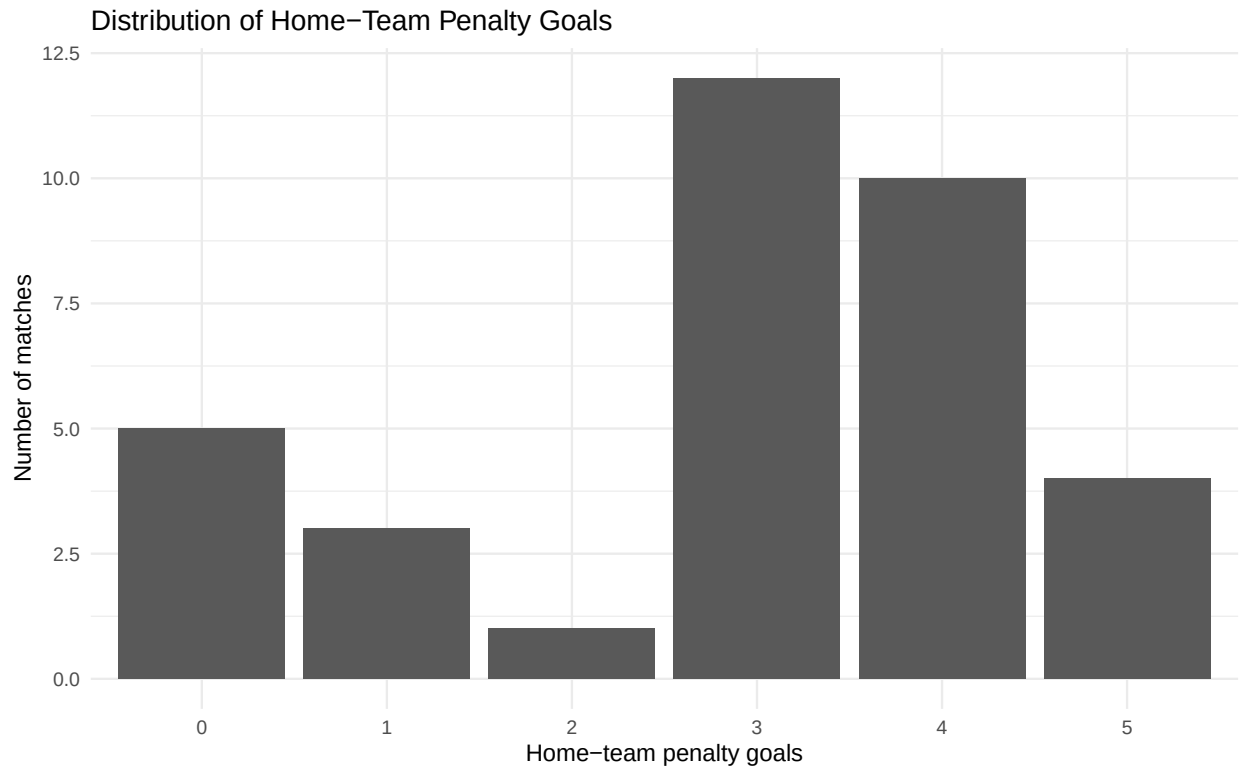
Results and Interpretation

```
# Keep only matches involving a penalty shootout.  
penalty_matches <- df %>%  
  filter(PenaltyShootout == 1)  
  
# Create a frequency table for HomePenalty.  
penalty_table <- table(penalty_matches$HomePenalty)  
penalty_table
```

```
##  
##  0  1  2  3  4  5  
##  5  3  1 12 10  4
```

```
# Find the most common HomePenalty value.  
most_common_penalty <- names(penalty_table)[which.max(penalty_table)]  
most_common_count <- max(penalty_table)
```

```
ggplot(penalty_matches, aes(x = factor(HomePenalty))) +  
  geom_bar() +  
  labs(  
    title = "Distribution of Home-Team Penalty Goals",  
    x = "Home-team penalty goals",  
    y = "Number of matches"  
  ) +  
  theme_minimal()
```



There were 35 matches involving a penalty shootout. The most common value of `HomePenalty` was 3, and it occurred in 12 matches. The taller bars show the penalty scores that occurred most often, while shorter bars represent less common scores.

Question 2: Distribution of Home Team Scores

Rationale

HomeTeamScore is a numerical variable, so histograms were used to show its distribution. A bin width of one was selected because football scores are whole numbers. The distributions were compared first by Stage, then by TournamentName, and finally by the three required AwayTeamScore groups.

The `aggregate()` function was used to calculate simple group means and medians. This provides numerical information to support the patterns shown in the histograms.

Distribution by Competition Stage

```
# Calculate the mean and median HomeTeamScore for each stage.
stage_mean <- aggregate(
  HomeTeamScore ~ Stage,
  mean,
  data = df
)

stage_median <- aggregate(
  HomeTeamScore ~ Stage,
  median,
  data = df
)

stage_mean
```

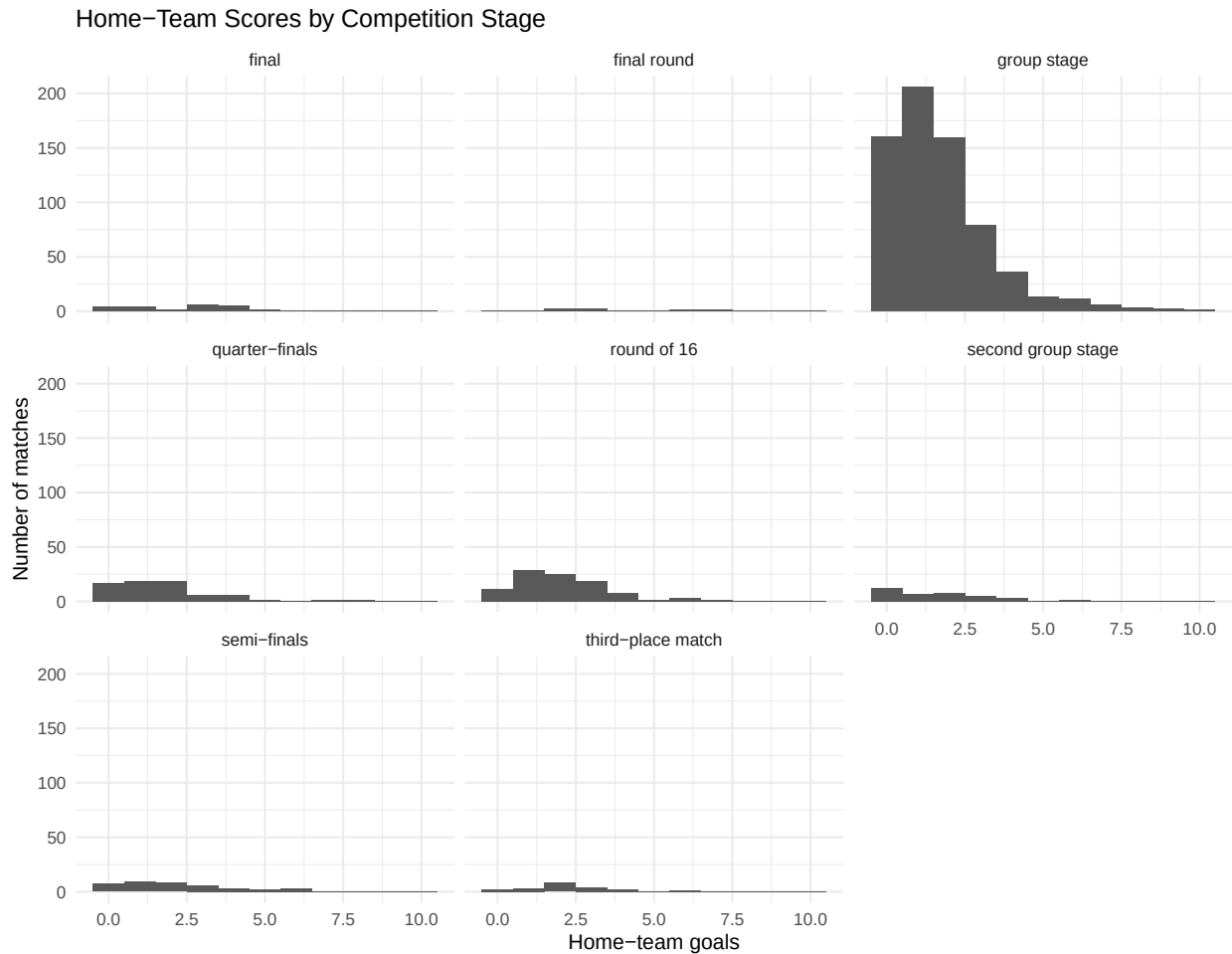
```
##           Stage HomeTeamScore
## 1           final      2.333333
## 2    final round      3.833333
## 3    group stage      1.671598
## 4 quarter-finals      1.700000
## 5    round of 16      2.041237
## 6 second group stage      1.555556
## 7    semi-finals      2.184211
## 8 third-place match      2.250000
```

```
stage_median
```

```
##           Stage HomeTeamScore
## 1           final              3
## 2    final round              3
## 3    group stage              1
## 4 quarter-finals              1
## 5    round of 16              2
## 6 second group stage          1
## 7    semi-finals              2
## 8 third-place match          2
```



```
ggplot(df, aes(x = HomeTeamScore)) +
  geom_histogram(binwidth = 1) +
  facet_wrap(~ Stage) +
  labs(
    title = "Home-Team Scores by Competition Stage",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Distribution by Tournament Name

```
# Calculate the mean and median HomeTeamScore for each tournament.
tournament_mean <- aggregate(
  HomeTeamScore ~ TournamentName,
  mean,
  data = df
)

tournament_median <- aggregate(
```

```

HomeTeamScore ~ TournamentName,
median,
data = df
)

```

tournament_mean

```

##      TournamentName HomeTeamScore
## 1  1930 FIFA World Cup      3.277778
## 2  1934 FIFA World Cup      2.823529
## 3  1938 FIFA World Cup      3.388889
## 4  1950 FIFA World Cup      3.136364
## 5  1954 FIFA World Cup      4.192308
## 6  1958 FIFA World Cup      2.514286
## 7  1962 FIFA World Cup      2.156250
## 8  1966 FIFA World Cup      2.156250
## 9  1970 FIFA World Cup      2.218750
## 10 1974 FIFA World Cup      1.342105
## 11 1978 FIFA World Cup      1.894737
## 12 1982 FIFA World Cup      1.846154
## 13 1986 FIFA World Cup      1.384615
## 14 1990 FIFA World Cup      1.230769
## 15 1994 FIFA World Cup      1.596154
## 16 1998 FIFA World Cup      1.531250
## 17 2002 FIFA World Cup      1.390625
## 18 2006 FIFA World Cup      1.343750
## 19 2010 FIFA World Cup      1.187500
## 20 2014 FIFA World Cup      1.265625
## 21 2018 FIFA World Cup      1.421875
## 22 2022 FIFA World Cup      1.578125

```

tournament_median

```

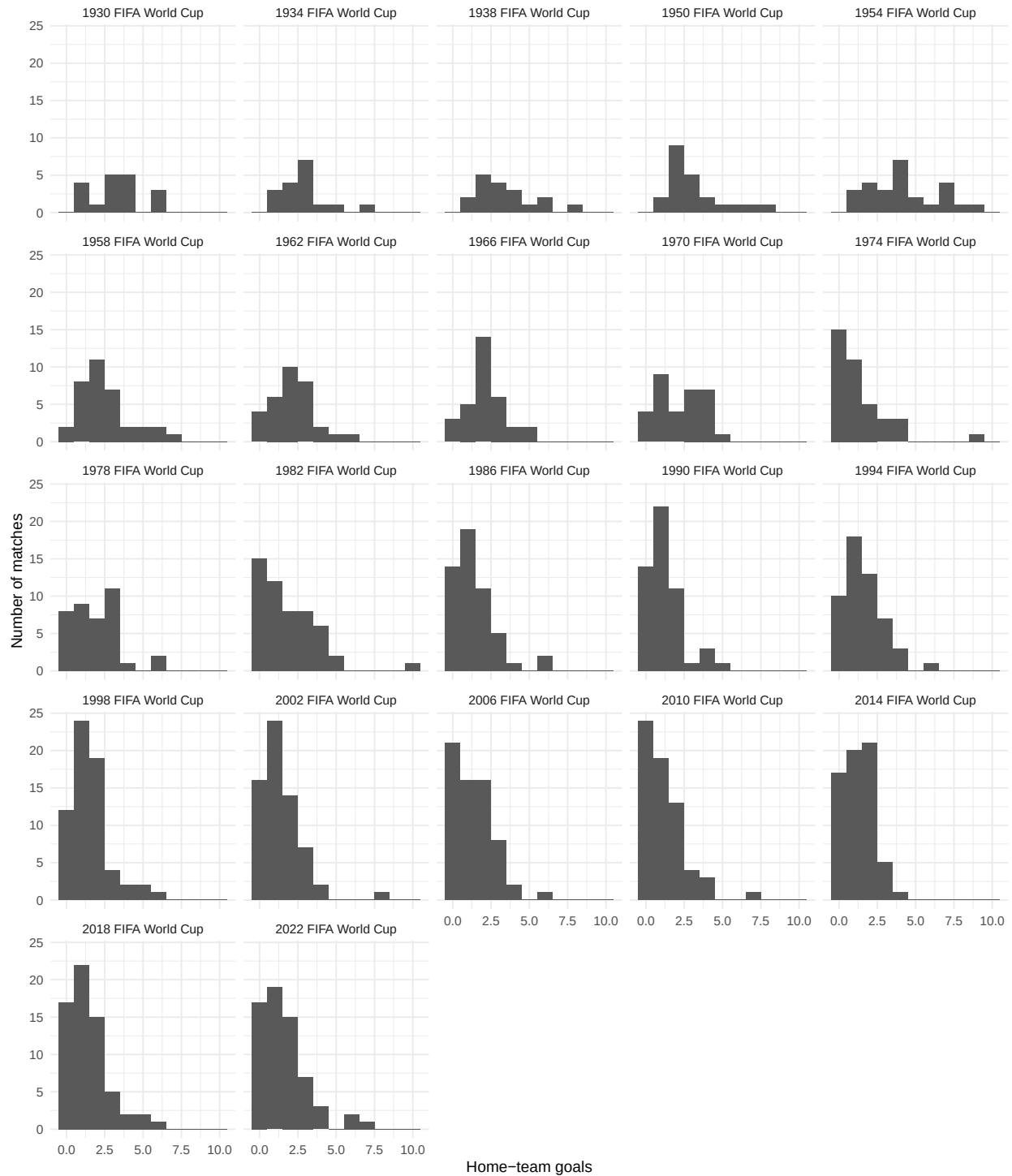
##      TournamentName HomeTeamScore
## 1  1930 FIFA World Cup          3.0
## 2  1934 FIFA World Cup          3.0
## 3  1938 FIFA World Cup          3.0
## 4  1950 FIFA World Cup          2.5
## 5  1954 FIFA World Cup          4.0
## 6  1958 FIFA World Cup          2.0
## 7  1962 FIFA World Cup          2.0
## 8  1966 FIFA World Cup          2.0
## 9  1970 FIFA World Cup          2.0
## 10 1974 FIFA World Cup          1.0
## 11 1978 FIFA World Cup          2.0
## 12 1982 FIFA World Cup          1.0
## 13 1986 FIFA World Cup          1.0
## 14 1990 FIFA World Cup          1.0
## 15 1994 FIFA World Cup          1.0
## 16 1998 FIFA World Cup          1.0
## 17 2002 FIFA World Cup          1.0
## 18 2006 FIFA World Cup          1.0
## 19 2010 FIFA World Cup          1.0

```

```
## 20 2014 FIFA World Cup      1.0
## 21 2018 FIFA World Cup      1.0
## 22 2022 FIFA World Cup      1.0
```

```
ggplot(df, aes(x = HomeTeamScore)) +  
  geom_histogram(binwidth = 1) +  
  facet_wrap(~ TournamentName) +  
  labs(  
    title = "Home-Team Scores by Tournament",  
    x = "Home-team goals",  
    y = "Number of matches"  
  ) +  
  theme_minimal()
```

Home-Team Scores by Tournament



Distribution by Away-Team Score Group

```
# Create an empty variable for the away-team score groups.
df$AwayScoreGroup <- NA
```

```

# Assign each match to one of the three required groups.
df$AwayScoreGroup[df$AwayTeamScore <= 1] <- "0-1 goals"

df$AwayScoreGroup[
  df$AwayTeamScore >= 2 &
  df$AwayTeamScore <= 3
] <- "2-3 goals"

df$AwayScoreGroup[df$AwayTeamScore >= 4] <- "4 or more goals"

# Convert the new variable to a factor.
df$AwayScoreGroup <- as.factor(df$AwayScoreGroup)

# Check the number of matches in each group.
table(df$AwayScoreGroup)

```

```

##
##      0-1 goals      2-3 goals 4 or more goals
##      701          237          26

```

```

# Calculate the mean and median HomeTeamScore for each group.
away_group_mean <- aggregate(
  HomeTeamScore ~ AwayScoreGroup,
  mean,
  data = df
)

away_group_median <- aggregate(
  HomeTeamScore ~ AwayScoreGroup,
  median,
  data = df
)

away_group_mean

```

```

##      AwayScoreGroup HomeTeamScore
## 1      0-1 goals      1.774608
## 2      2-3 goals      1.746835
## 3 4 or more goals      1.730769

```

```

away_group_median

```

```

##      AwayScoreGroup HomeTeamScore
## 1      0-1 goals      1
## 2      2-3 goals      1
## 3 4 or more goals      1

```

```

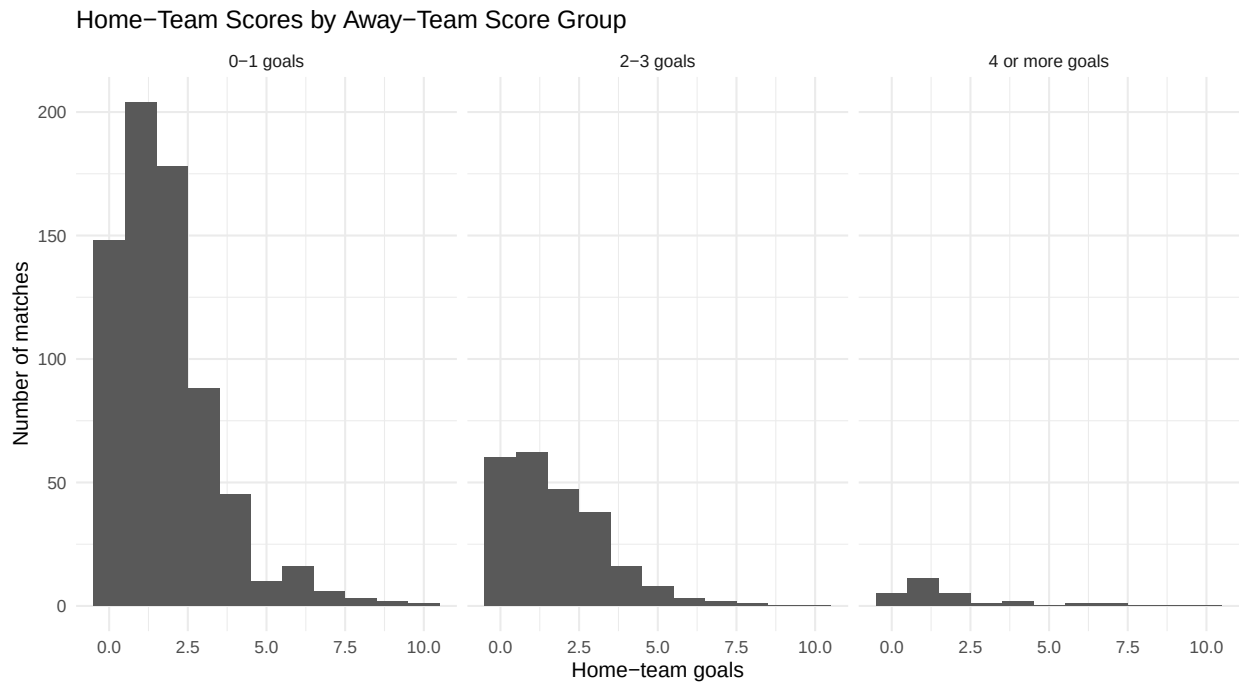
ggplot(df, aes(x = HomeTeamScore)) +
  geom_histogram(binwidth = 1) +
  facet_wrap(~ AwayScoreGroup) +
  labs(

```

```

title = "Home-Team Scores by Away-Team Score Group",
x = "Home-team goals",
y = "Number of matches"
) +
theme_minimal()

```



Code Testing

```

# Check the score variable types.
class(df$HomeTeamScore)

```

```
## [1] "numeric"
```

```
class(df$AwayTeamScore)
```

```
## [1] "numeric"
```

```

# Check for missing score values.
sum(is.na(df$HomeTeamScore))

```

```
## [1] 0
```

```
sum(is.na(df$AwayTeamScore))
```

```
## [1] 0
```

```
# Check the new group variable.
class(df$AwayScoreGroup)
```

```
## [1] "factor"
```

```
table(df$AwayScoreGroup)
```

```
##
##      0-1 goals      2-3 goals 4 or more goals
##           701           237           26
```

```
sum(is.na(df$AwayScoreGroup))
```

```
## [1] 0
```

```
# Check that all matches are included in the three groups.
sum(table(df$AwayScoreGroup))
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

HomeTeamScore and AwayTeamScore should be numeric variables; R may display their class as `integer` because football goals are whole numbers. The missing-value checks should return zero. `AwayScoreGroup` should be a factor, and the total number shown by its frequency table should equal the total number of rows in `df`.

Results and Interpretation

The stage with the highest average home-team score was **final round**, with an average of 3.83 goals. The lowest average occurred in **second group stage**, with 1.56 goals. The stage histograms also show whether the scores were concentrated at low values or spread across a wider range.

The tournament with the highest average home-team score was **1954 FIFA World Cup**, with 4.19 goals. The lowest tournament average occurred in **2010 FIFA World Cup**, with 1.19 goals. The tournament histograms make it possible to compare how the score distributions changed between World Cup editions.

Among the three away-team score groups, **0-1 goals** had the highest average home-team score at 1.77 goals. This result describes a relationship in the dataset, but it does not prove that the away-team score caused the home-team score.

Question 3: Penalty Shootout Participation

Rationale

Only matches where `PenaltyShootout` equals 1 were selected because this question focuses on matches decided by a penalty shootout. The matches were grouped separately by home team and away team because the assignment requires two different rankings.

The number of matches for each team was counted using `group_by()` and `summarise()`. The results were sorted from highest to lowest, and `head()` was used to select the first five teams.

Filter Penalty Shootout Matches

```
# Keep only matches that involved a penalty shootout.
penalty_matches <- df %>%
  filter(PenaltyShootout == 1)
```

```
# Check the number of penalty-shootout matches.
nrow(penalty_matches)
```

```
## [1] 35
```

```
# Check that the filtered data contains only value 1.
table(penalty_matches$PenaltyShootout)
```

```
##
##  1
## 35
```

Top Five Home Teams

```
# Group the penalty matches by home-team name and code.
# Count the number of matches for each home team.
home_penalty_count <- penalty_matches %>%
  group_by(HomeTeamName, HomeTeamCode) %>%
  summarise(NumberOfMatches = n())

# Sort the teams from the highest count to the lowest count.
home_penalty_count <- home_penalty_count %>%
  arrange(desc(NumberOfMatches))

# Select the first five teams.
top5_home <- head(home_penalty_count, 5)

# Change the column names to match the assignment requirement.
names(top5_home) <- c(
  "TeamName",
  "TeamCode",
  "NumberOfMatches"
```



```
)

# Display the top five home teams as a table.
top5_home
```

```
## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName      TeamCode NumberOfMatches
##   <chr>         <chr>          <int>
## 1 Argentina     ARG                4
## 2 Brazil        BRA                4
## 3 Spain         ESP                4
## 4 Netherlands  NLD                3
## 5 West Germany DEU                3
```

Top Five Away Teams

```
# Group the penalty matches by away-team name and code.
# Count the number of matches for each away team.
away_penalty_count <- penalty_matches %>%
  group_by(AwayTeamName, AwayTeamCode) %>%
  summarise(NumberOfMatches = n())

# Sort the teams from the highest count to the lowest count.
away_penalty_count <- away_penalty_count %>%
  arrange(desc(NumberOfMatches))

# Select the first five teams.
top5_away <- head(away_penalty_count, 5)

# Change the column names to match the assignment requirement.
names(top5_away) <- c(
  "TeamName",
  "TeamCode",
  "NumberOfMatches"
)

# Display the top five away teams as a table.
top5_away
```

```
## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName      TeamCode NumberOfMatches
##   <chr>         <chr>          <int>
## 1 France       FRA                5
## 2 Argentina    ARG                3
## 3 England      ENG                3
## 4 Croatia      HRV                2
## 5 Italy        ITA                2
```

Code Testing

```
# Check that all filtered matches involved a penalty shootout.  
table(penalty_matches$PenaltyShootout)
```

```
##  
## 1  
## 35
```

```
# Both top-five tables should contain five rows.  
nrow(top5_home)
```

```
## [1] 5
```

```
nrow(top5_away)
```

```
## [1] 5
```

```
# Check that both tables have the required column names.  
names(top5_home)
```

```
## [1] "TeamName"      "TeamCode"      "NumberOfMatches"
```

```
names(top5_away)
```

```
## [1] "TeamName"      "TeamCode"      "NumberOfMatches"
```

```
# Check for missing values in team names and codes.  
sum(is.na(top5_home$TeamName))
```

```
## [1] 0
```

```
sum(is.na(top5_home$TeamCode))
```

```
## [1] 0
```

```
sum(is.na(top5_away$TeamName))
```

```
## [1] 0
```

```
sum(is.na(top5_away$TeamCode))
```

```
## [1] 0
```

```
# Check that the match counts are sorted from high to low.
top5_home$NumberOfMatches
```

```
## [1] 4 4 4 3 3
```

```
top5_away$NumberOfMatches
```

```
## [1] 5 3 3 2 2
```

The `PenaltyShootout` table should contain only the value 1. Both top-five tables should contain five rows and the columns `TeamName`, `TeamCode`, and `NumberOfMatches`. The missing-value checks should return zero. The match counts should be displayed from the highest value to the lowest value.

Results and Interpretation

The national team ranked first in the home-team table was **Argentina**, which participated in 4 penalty-shootout matches as the home team. The national team ranked first in the away-team table was **France**, which participated in 5 matches as the away team.

There were 1 teams appearing in both top-five lists: **Argentina**. The rankings were **different** when the team names and their order were compared.

The difference between the two rankings may occur because teams can be recorded as the home team in some matches and the away team in other matches. These labels do not necessarily mean that the match was played in the team own country.

Question 4: Factors Influencing Match Outcomes

Selected Factors and Rationale

The two selected factors were `Stage` and `ExtraTime`.

`Stage` was selected because match conditions may differ between group-stage matches and knockout-stage matches. For example, teams may use different strategies in finals, semi-finals, or group-stage matches.

`ExtraTime` was selected because matches that continue beyond regular time may have a different distribution of results compared with matches that finish during regular time.

Frequency tables were used to show the number of home-team wins, away-team wins, and draws. Percentage tables were also created because the numbers of matches in the different groups are not equal. A proportional stacked bar chart was selected because it allows the three result categories to be compared within each group.

Overall Match Results

```
# Count the number of matches in each Result category.
overall_result_table <- table(df$Result)

overall_result_table
```

```
##
## away team win      draw home team win
##           240           179           545
```

```
# Find the most common match result.
most_common_result <- names(overall_result_table)[
  which.max(overall_result_table)
]

most_common_result_count <- max(overall_result_table)
```

Factor 1: Competition Stage

```
# Create a table of Stage and Result.
stage_result_count <- table(
  df$Stage,
  df$Result
)

stage_result_count
```

```
##
##           away team win draw home team win
## final           4      0           17
## final round       0      1           5
## group stage     164    166          346
## quarter-finals   24      2           44
```

##	round of 16	21	2	74
##	second group stage	11	8	17
##	semi-finals	10	0	28
##	third-place match	6	0	14

```

# Convert the counts into row percentages.
# margin = 1 calculates percentages within each Stage.
stage_result_percent <- prop.table(
  stage_result_count,
  margin = 1
) * 100

# Round the percentages to one decimal place.
stage_result_percent <- round(
  stage_result_percent,
  1
)

stage_result_percent

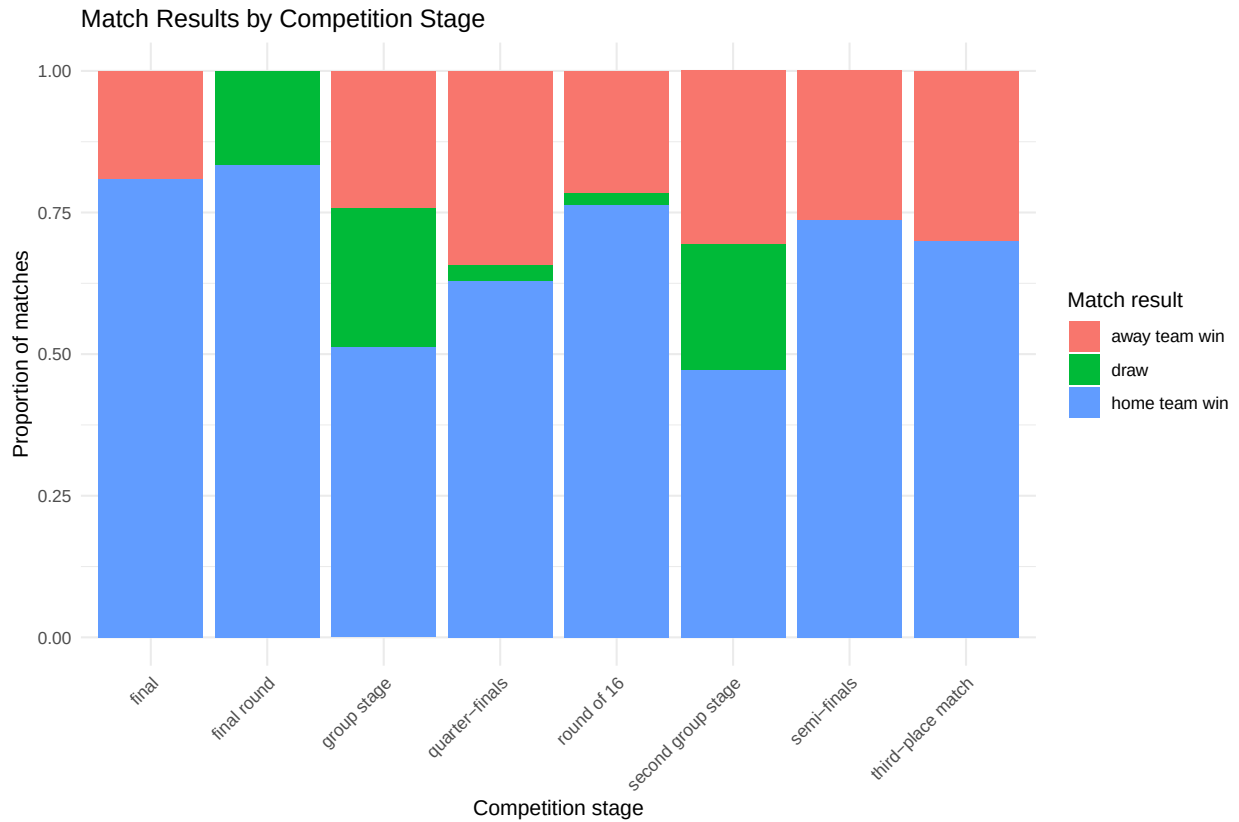
```

##		away team win	draw	home team win
##	final	19.0	0.0	81.0
##	final round	0.0	16.7	83.3
##	group stage	24.3	24.6	51.2
##	quarter-finals	34.3	2.9	62.9
##	round of 16	21.6	2.1	76.3
##	second group stage	30.6	22.2	47.2
##	semi-finals	26.3	0.0	73.7
##	third-place match	30.0	0.0	70.0

```

# Create a proportional bar chart of Result for each Stage.
ggplot(df, aes(x = Stage, fill = Result)) +
  geom_bar(position = "fill") +
  labs(
    title = "Match Results by Competition Stage",
    x = "Competition stage",
    y = "Proportion of matches",
    fill = "Match result"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  )

```



Factor 2: Extra Time

Create a table of ExtraTime and Result.

```
extra_time_result_count <- table(
  df$ExtraTime,
  df$Result
)
```

```
extra_time_result_count
```

```
##
##      away team win draw home team win
##    0          215  173          503
##    1           25    6           42
```

Convert the counts into row percentages.

margin = 1 calculates percentages within each ExtraTime group.

```
extra_time_result_percent <- prop.table(
  extra_time_result_count,
  margin = 1
) * 100
```

Round the percentages to one decimal place.

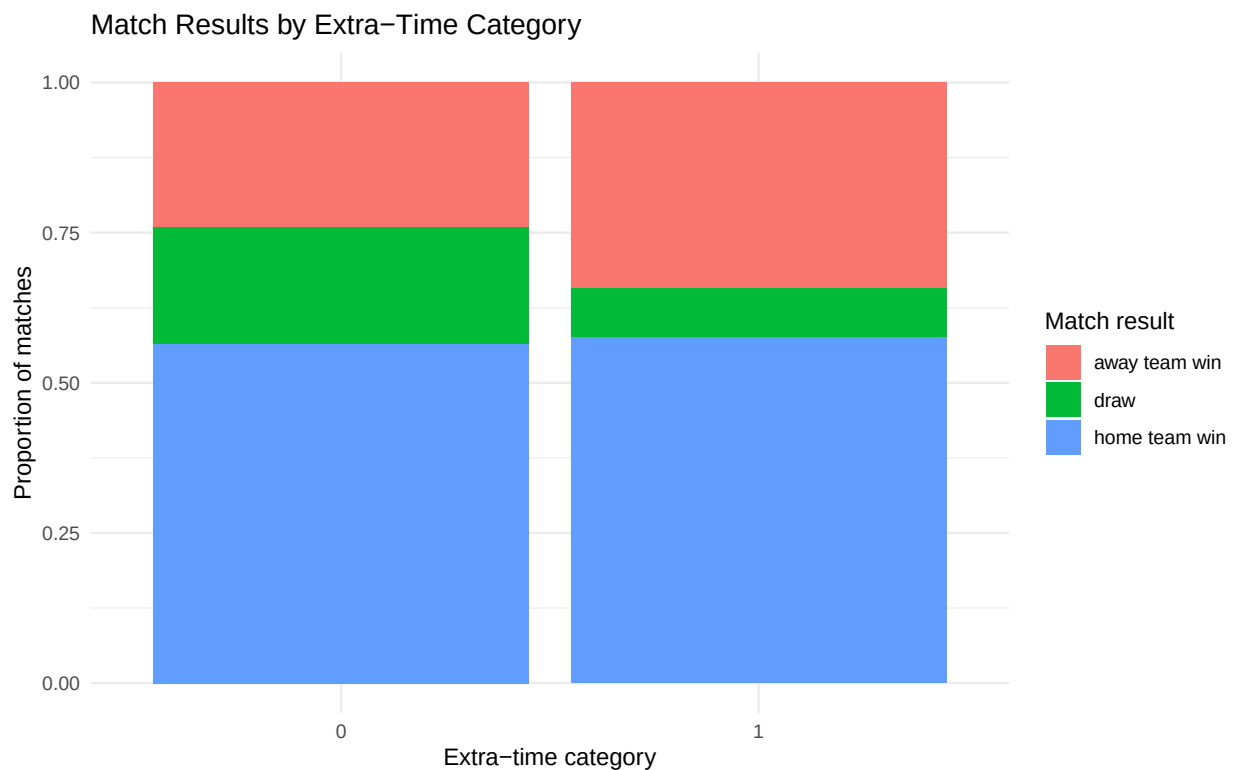
```
extra_time_result_percent <- round(
```

```
extra_time_result_percent,
1
)

extra_time_result_percent
```

```
##
##      away team win draw home team win
##    0          24.1 19.4          56.5
##    1          34.2  8.2          57.5
```

```
# Create a proportional bar chart of Result by ExtraTime.
ggplot(df, aes(x = ExtraTime, fill = Result)) +
  geom_bar(position = "fill") +
  labs(
    title = "Match Results by Extra-Time Category",
    x = "Extra-time category",
    y = "Proportion of matches",
    fill = "Match result"
  ) +
  theme_minimal()
```



Code Testing

```

# Check that the three variables are factors.
class(df$Result)

## [1] "factor"

class(df$Stage)

## [1] "factor"

class(df$ExtraTime)

## [1] "factor"

# Display the categories in each variable.
levels(df$Result)

## [1] "away team win" "draw" "home team win"

levels(df$Stage)

## [1] "final" "final round" "group stage"
## [4] "quarter-finals" "round of 16" "second group stage"
## [7] "semi-finals" "third-place match"

levels(df$ExtraTime)

## [1] "0" "1"

# Check for missing values.
sum(is.na(df$Result))

## [1] 0

sum(is.na(df$Stage))

## [1] 0

sum(is.na(df$ExtraTime))

## [1] 0

# The total number in each table should equal the number of rows in df.
sum(stage_result_count)

## [1] 964

```



```
nrow(df)
```

```
## [1] 964
```

```
sum(extra_time_result_count)
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```
# Each row of the percentage tables should total approximately 100%.
```

```
rowSums(stage_result_percent)
```

```
##           final           final round           group stage           quarter-finals
##           100.0           100.0           100.1           100.1
## round of 16 second group stage           semi-finals third-place match
##           100.0           100.0           100.0           100.0
```

```
rowSums(extra_time_result_percent)
```

```
##      0      1
## 100.0  99.9
```

`Result`, `Stage`, and `ExtraTime` should return `"factor"`. The missing-value checks should return zero. The totals from the two count tables should equal the total number of rows in `df`. Each row of the percentage tables should total approximately 100%. Small differences such as 99.9% or 100.1% may occur because the percentages were rounded.

Results and Interpretation

Across all matches, the most common result was **home team win**, which occurred in 545 matches.

For competition stage, **final round** had the highest proportion of home-team wins, at 83.3%. The percentage table and proportional bar chart show that the distribution of home-team wins, away-team wins, and draws was not identical across all competition stages. This suggests that **Stage** is associated with differences in match outcomes.

For extra time, category **1** had the highest proportion of home-team wins, at 57.5%. The percentage table and bar chart allow the result distribution for matches with different extra-time categories to be compared.

These findings show associations in the FIFA World Cup dataset. They do not prove that competition stage or extra time directly caused a particular match result.

Question 5: Git and GitHub

GitHub Repository

Git and GitHub were used to manage the development of this project.

The GitHub repository is available at:

<https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis>

Git was used to record changes made to the project. GitHub was used to store the repository online and maintain a history of the project development.

Project Folder Structure

The project was organised into separate folders:

- **data/** contains the four FIFA World Cup datasets.
- **scripts/** contains the R scripts used for Questions 1 to 4.
- **report/** contains the R Markdown report.
- **figures/** contains images used in the report.
- **COMP1013-FIFA-WorldCup-Analysis.Rproj** is the RStudio project file.

This folder structure keeps the datasets, scripts, report and figures separate, making the project easier to understand and maintain.

Commit History

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis$ git log --oneline --decorate -10
ab15faa (HEAD -> main, origin/main, origin/HEAD) Complete Questions 3 and 4 analysis
c30d58a Fix report paths and update Questions 1 and 2
d787645 Change path of datasets
b0b1751 Simplify code and complete Questions 1 and 2
5abea0e Complete Questions 1 and 2 data analysis
2006d5f Complete Question 1 data preparation and penalty analysis
cb5ebd3 Add FIFA World Cup datasets
c29a184 Set up RStudio project and report template
6cf6ab0 Create initial project structure
e273768 Initial commit
```

Explanation of the Commit History

The commit history shows that the project was developed gradually through several meaningful commits.

1. Create initial project structure

This commit created the main folders needed for the project.

2. **Set up RStudio project and report template**
This commit added the RStudio project file and the initial R Markdown report.
3. **Add FIFA World Cup datasets**
This commit added the four CSV datasets used in the analysis.
4. **Complete Question 1 data preparation and penalty analysis**
This commit added the data preparation and penalty analysis for Question 1.
5. **Complete Questions 1 and 2 data analysis**
This commit added and improved the analyses for Questions 1 and 2.
6. **Fix report paths and update Questions 1 and 2**
This commit corrected the dataset paths and updated the report content.
7. **Complete Questions 3 and 4 analysis**
This commit added the penalty-shootout rankings and match-outcome analysis.

The commit messages describe the purpose of each change and demonstrate the progress of the project over time.